JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 93, NO. B4, PAGES 3387–3400, APRIL 10, 1988

Multicomponent Paleomagnetic Data From the Nosoni Formation, Eastern Klamath Mountains, California: Cratonic Permian Primary Directions With Jurassic Overprints

PAUL R. RENNE[1]

Department of Geology and Geophysics, University of California, Berkeley

GARY R. SCOTT

Lodestar Magnetics Inc., Oakland, California

DAVID R. BAZARD

Department of Geosciences, University of Arizona, Tuscon

Paleomagnetic data from the Guadelupian (Upper Permian) Nosoni Formation of the Redding Section (RS) consistently exhibit two directional components of magnetization in addition to a late Quaternary normal weathering overprint. Six sites in ash flow tuffs and tuffaceous sedimentary rocks, spanning 200 m of section, yield data that allow precise characterization of all components. Two additional sites in calcareous mudstones were too heavily overprinted to yield stable paleomagnetic directions. Thermal demagnetization of the tuffaceous samples in 16–18 steps between natural remanent magnetization and 660°C provides a detailed data base from which to calculate directional components. Analysis of demagnetization trends identifies a prefolding reversed direction carried by both magnetite and hematite and a postfolding normal direction carried by hematite only. Petrologic considerations suggest paragenesis of these oxides in the Permian (primary magnetite and hematite) and Early–Middle Jurassic (metamorphic hematite). Least squares analysis of progressively demagnetized vector endpoint trends between 540° and 600°C yields a formation mean direction in magnetite having $D = 156.4°$, $I = -29.7°$, $k = 101$, $\alpha_{95} = 7.6°$. This result is in close agreement with the composite prefolding direction ($D = 160.3°$, $I = -31.5°$), carried by both magnetite and hematite, determined by the intersection of remagnetization circles. The postfolding hematite-carried overprint direction calculated by the latter technique has $D = 352.3°$, $I = 35.0°$. Paleopoles corresponding to these two directions are located at 57.6°N, 103.0°E (magnetite formation mean) and 67.0°N, 104.1°E (hematite overprint), coinciding within error to Upper Permian and Early Jurassic (respectively) reference poles from the North American craton. These data indicate no significant rotation or latitudinal translation of the sample sites with respect to cratonic North America since the Permian. This conclusion contrasts with seemingly reliable paleomagnetic results from elsewhere in the RS, which indicate major post-Permian clockwise rotations. These rotations are evidently specific to individual structural domains and are probably manifestations of oroclinal bending about vertical axes.

INTRODUCTION

Geology of the Redding Section

The Redding Section (RS) of the eastern Klamath Mountains consists of weakly metamorphosed and moderately deformed Devonian through Jurassic strata, mostly of volcanic arc derivation [*Murray and Condie*, 1973; *LaPierre et al.*, 1985; P. R. Renne and M. M. Miller, manuscript in preparation, 1988]. The RS is the easternmost tectonostratigraphic terrane of the Klamath Mountains Province (Figure 1) and is characterized internally by a coherent stratigraphy, with conformable or paraconformable depositional contacts between the 17 recognized Paleozoic and Mesozoic formations [*Renne and Scott*, 1988]. The structure of the RS is a complex superposition of two distinct deformation events. The principal deformation (F_1) produced open to isoclinal folds and penetrative foliation and is constrained to be of Middle Jurassic age by K-Ar dates on postfolding instrusions of 168 and 166 Ma [*Renne et al.*, 1986*a*; *Renne and Scott*, 1988] and by the Bajocian biostratigraphic age [*Sanborn*, 1960] of the youngest folded strata. Formational contacts and F_1 structures (fold axes and cleavage) define a sinuous pattern of variable orientations; NW structural trends are dominant in the central part of the region, whereas the northern and southern portions are characterized by NE to E-W trends. This variation in the orientation of F_1 structures appears to be the result of oroclinal deformation (F_2) [*Renne et al.*, 1986*a*; *Renne and Scott*, 1986; *Renne and Scott*, 1988], which is of post-Middle Jurassic age.

Plate Tectonic Models

The RS is among many tectonostratigraphic terranes of the western North American Cordillera which are considered by many to be "suspect terranes" [*Coney et al.*, 1980] because their relationship to each other and to the North American craton is uncertain. The possibility exists that the RS and other suspect terranes originated far from the continental

[1]Now at Department of Geological and Geophysical Sciences, Princeton University, Princeton, New Jersey.

Copyright 1988 by the American Geophysical Union.

Paper number 7B3019.
0148-0227/88/007B-3019$05.00

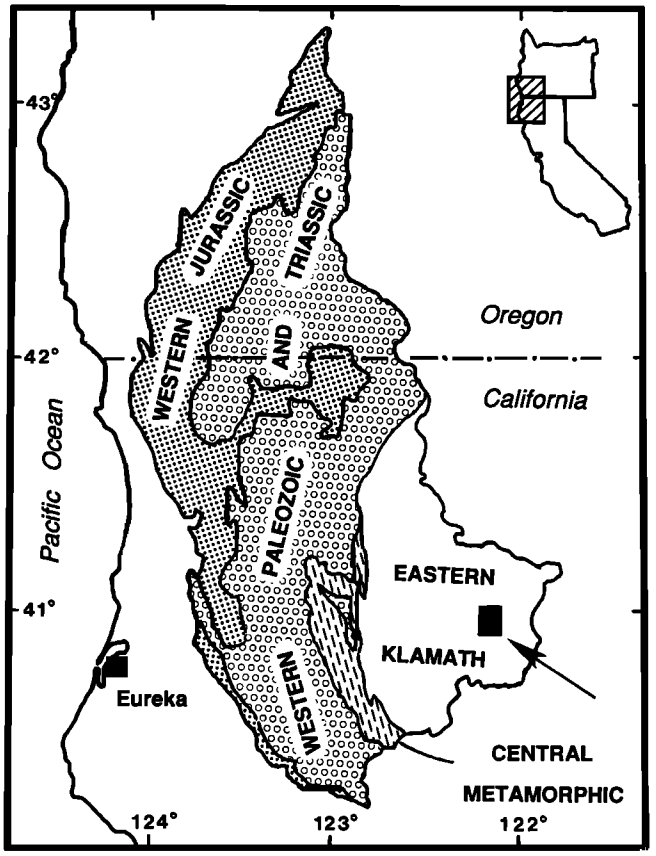


Fig. 1. Location map showing distribution of lithotectonic belts (terranes) of the Klamath Mountains. Area of Figure 2 shown as rectangle.

margin, having been accreted to it subsequently through the consumption of intervening oceanic lithosphere during subduction. Unconstrained by details of the stratigraphy, petrology, structure, paleontology, and paleomagnetism of the RS, many have incorporated this terrane into diverse and mutually inconsistent models of Cordilleran tectonic evolution (e.g., compare the conclusions of *Wright* [1982] with those of *Speed* [1979]). Controversy exists about virtually every aspect of the role of the RS in these various models: whether or not the RS is allochthonous with respect to North America, the polarity of subduction during its genesis, and so on.

Several paleomagnetic studies [*Achache et al.*, 1982; *Fagin and Gose*, 1983; *Mankinen et al.*, 1984; *Scott et al.*, 1985; *Renne et al.*, 1986*b*] have been undertaken in the RS in attempts to constrain the plate tectonic history of the terrane. These studies concur in reporting no significant latitudinal displacement of the RS relative to the North American craton, but substantially different (0–116°) amounts of clockwise rotation have been inferred. Major differences exist between the mechanisms of rotation proposed in these studies, and disagreement exists as to the exact timing of rotation. This paper presents new paleomagnetic data which combined with concurrent mapping, structural, and geochronologic study of the RS place further constraints on the mechanism and timing of tectonic rotations manifest in this terrane.

NOSONI FORMATION

Data reported in this paper were obtained from eight sites in the Permian Nosoni Formation, all located within the McCloud River drainage in the Bollibokka 15 arc min quadrangle and shown in Figure 2. The Nosoni Formation contains well-characterized fusulinid, brachiopod, and coral faunas of Guadelupian (late Middle Permian) stage age [*Coogan*, 1960; *Albers and Robertson*, 1961; *Skinner and Wilde*, 1965]. The Nosoni Formation consists mainly of volcaniclastic rocks, including tuffs, volcanic breccias, tuffaceous sandstones, and sparse lavas. Volumetrically minor mudstones and siltstones are intercalated throughout the section and are locally fossiliferous.

The stratigraphic and sedimentologic character of Nosoni sediments indicate derivation from nearby volcanic sources. The petrologic/geochemical character of Nosoni volcanics suggests that this volcanism occurred in a primitive island arc setting [*Murray and Condie*, 1973; P. R. Renne and M. M. Miller, manuscript in preparation, 1988]. Nosoni faunas provide evidence for shallow marine depositional environments [*Coogan*, 1960], and the presence of columnar-jointed lavas and densely welded ash flow tuffs indicate occasional emergence into a subaerial environment.

The general down-section increase in grade of metamorphic assemblages in the RS, from zeolite facies in the Jurassic Potem Formation through greenschist facies in the Devonian Copley Greenstone, is typical of burial metamorphism under an intermediate geothermal gradient [*Renne and Scott*, 1988]. Secondary mineral assemblages developed in Nosoni rocks include prehnite, pumpellyite, albite, chlorite, sphene, and quartz, indicative of the prehnite-pumpellyite facies [*Turner*, 1981]. Consideration of metamorphic phase equilibria indicates that the Nosoni Formation in the area of this study was subjected to maximum temperatures between 275° and 325°C during burial metamorphism [*Renne and Scott*, 1988]. Following F_1 deformation in the Early to Middle Jurassic, regional cooling of the RS was interrupted only locally by Middle Jurassic and Early Cretaceous hypabyssal intrusions [*Renne and Scott*, 1988] that occur 8 km and farther from the sample sites described below. Hydrothermal activity is manifested in the Kuroko-type massive sulfide ore deposits of the East Shasta copper-zinc district [*Albers and Robertson*, 1961]. These deposits are hosted by the Bully Hill Rhyolite and Pit Formation, which overlie the Nosoni Formation; thus Nosoni rocks were also subjected to local hydrothermal activity, represented by quartz veins and disseminated pyrite found in them.

SAMPLE SITES

Sample site locations are shown in Figure 2. Each site represents a single lithologic unit except for site 14, where two different depositional units were sampled. Sites were chosen to encompass a variety of rock types and structural attitudes. The sites are distributed over approximately 3.5 km along strike, thus permitting the recognition of possible localized effects such as hydrothermal alteration, proximity to intrusions, weathering, variable initial dip, slumping, or lightning strikes. A primary objective of this study was to acquire paleomagnetic data for which a fold test would be applicable because the thermal history of this region suggests the possibility of one or more overprinting events, whose ages must be constrained in order to permit meaningful tectonic interpretations. Sampling sites were chosen to maximize the variation of paleohorizontal attitudes, which are shown in Tables 1 and 3. Sampling was restricted to outcrops where bedding (or some other indication of paleohorizontal) was unambiguous and where this attitude could be measured directly on the outcrop.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

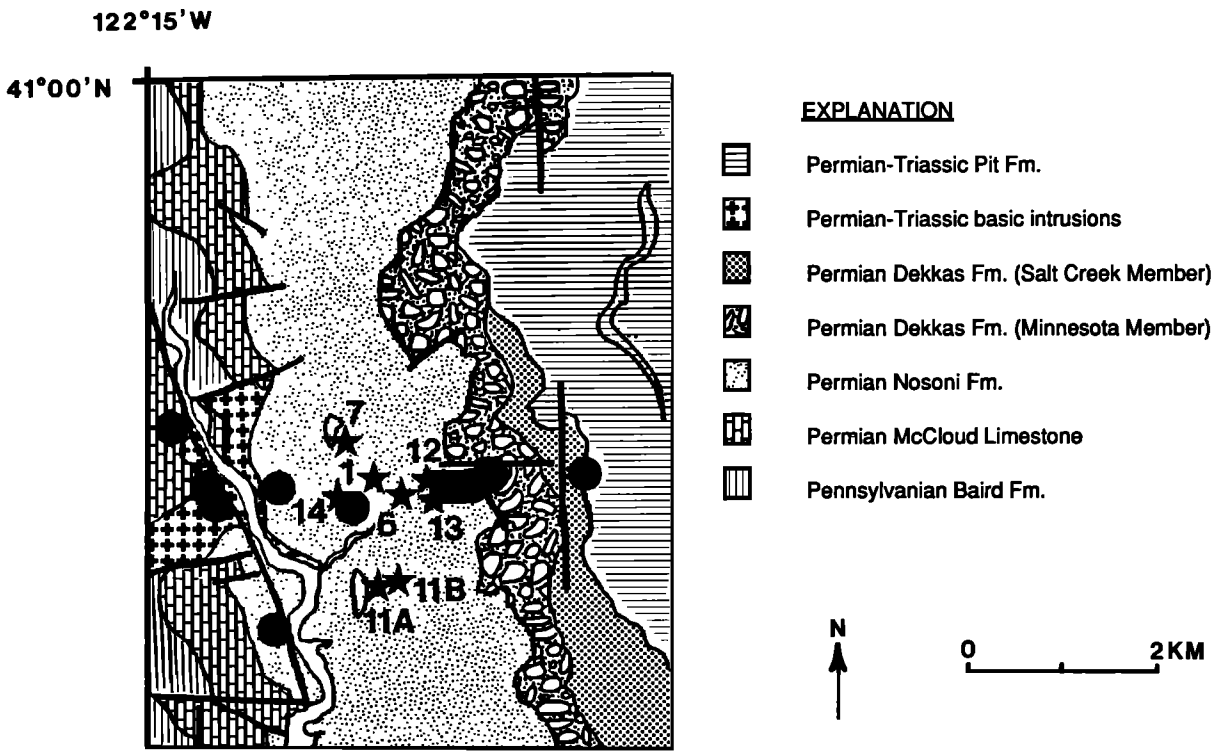


Fig. 2. Geologic map of a part of the Bollibokka 15 arc min quadrangle, showing paleomagnetic site localities. Sites of this study are denoted by numbered stars. Sites of *Mankinen et al.* [1984] are shown as solid circles; that of *Fagin and Gose* [1983] is shown as solid square. Unpatterned area is McCloud Arm of Shasta Lake; tributary near site 6 is Nosoni Creek.

Samples spanned the maximum exposed stratigraphic range of each sampled unit.

Sites N-1, 6, and 7 occur in the Skookum Tuff [*Scott et al.*, 1985], a densely welded dacitic ignimbrite. The Skookum Tuff is somewhat discontinuous along strike, ranging from 0 to 50 m thick, and appears to have been deposited onto a surface of appreciable relief. It is a dense, dark purplish-brown, extremely hard rock that possesses abundant light-colored relict fiamme. Fiamme define a eutaxitic foliation whose orientation parallels bedding in sediments above and below the Skookum; the attitude of Skookum sites was measured directly on this feature. In thin section, densely welded vitroclastic texture is spectacularly well preserved, as are pervasive coarse devitrification spherulites and axiolites which are now composed of quartz and albite. Skookum phenocrysts are solely of plagioclase (homogeneously albitized) and rare pseudomorphed mafic minerals (probably orthopyroxene or amphibole) now altered to chlorite and pumpellyite. Occasional lithic fragments comprise andesite and dacite. Microbreccia pipes containing abundant pyrite are visible in some outcrops; these may represent fossil fumaroles.

Site N-11A is in a tuff that resembles the Skookum in color, density, and extreme hardness but lacks fiamme and recognizable vitroclastic texture. Exposed thickness is about 30 m. Mesoscopic lamination is present, suggesting that this tuff is welded, though not so densely as the Skookum. The matrix of this tuff is completely recrystallized to quartz and albite, leaving no trace of primary texture. As with the Skookum, its sole phenocrysts are of albitized plagioclase and rare pseudomorphed mafic minerals, but in this tuff the lithic component

TABLE 1. Magnetite Direction Data

Site	Structural Attitude: Dip	Structural Attitude: Dip Direction	Temperature Range, °C	Number of Measurements	Geographic Coordinates, deg: Declination	Geographic Coordinates, deg: Inclination	Stratigraphic Coordinates, deg: Declination	Stratigraphic Coordinates, deg: Inclination	Statistics: N	Statistics: α_{95}	Statistics: k	Pole Position (Stratigraphic Coordinates): Latitude, °N	Pole Position (Stratigraphic Coordinates): Longitude, °E	Error: dp	Error: dm
1	30	111	540–590	3	147.9	3.6	150.6	−20.2	8	3.4	274	−50.0	−074.2	1.8	3.5
6	43	133	540–590	3	148.0	8.6	150.8	−32.8	11	4.1	124	−55.9	−066.7	2.6	4.7
7	82	068	540–590	3	121.3	−5.2	158.5	−37.0	7	6.8	79	−62.7	−073.7	4.7	8.0
11A	37	077	540–600	4	144.8	−22.8	163.6	−30.9	8	3.9	201	−61.9	−087.3	2.4	4.3
11B	37	077	540–600	4	144.1	−16.6	158.9	−26.9	11	4.2	119	−57.6	−081.7	2.5	4.6
Mean					156.4	−29.7			5	7.6	101	−57.6	−077.0	4.7	8.4

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(also of andesite and dacite fragments) is much more abundant.

Site 11B is in a massive crystal-lithic tuff that overlies the tuff of site 11A, separated from the latter by approximately 8 m of poorly exposed calcareous mudstone. Exposed thickness of the unit is about 10 m. This tuff is reddish-brown on weathered surfaces, but fresh surfaces reveal a greenish color largely devoid of oxidized pigment. The lithic component of this tuff comprises andesite and dacite fragments of diverse textures as large as 4 cm in diameter and composes more than 75% of the clastic grains. The crystals are of albitized plagioclase and minor altered mafic minerals. The matrix is wholly recrystallized to quartz and albite, leaving no trace of primary texture.

Site 14 is in two crystal-lithic tuffs whose lithologies are virtually identical and which directly overlie one another. Both are dark purplish brown and massive in outcrop, and both are approximately 5 m thick. Angular lithic fragments compose over 90% of the clastic grains, are composed of basalt, andesite and dacite, and range to 2 cm in diameter. Albitized plagioclase and traces of augite occur as angular crystals which may be either accidental or cognate. No vitroclastic texture is evident in the turbid matrix, which includes particles as small as fine mud. These units probably represent shallow water or subaerial debris flow deposits containing little or no primary magmatic constituents. This site is about 200 m stratigraphically below site 1.

Sites 12 and 13 are from well-bedded, fossiliferous, calcareous mudstones. Both sites are from the same sequence of sedimentary strata which occur in slightly different structural positions, separated by about 50 m. Samples are from different beds in the sequence, which is about 25 m thick. When fresh, these mudstones are dark bluish black, weathering to buff color with dissolution of the carbonate matrix. The abundant fusilinids, brachiopods, and corals from these localities are described by *Coogan* [1960]. These mudstones are all tuffaceous to some extent, containing angular grains of plagioclase and quartz.

Oxide Mineralogy and Petrography

Polished thin sections of samples from sites 1, 11A, and 11B were examined in an effort to characterize the oxide phases and aid in interpretation of the paleomagnetic data. In each sample, fine (up to about 10 μm diameter) anhedral oxide grains are disseminated throughout the tuffaceous matrix. These were identified as hematite (or titanohematite) on the basis of their reflectivity and their anisotropy in polarized reflected light.

Larger opaque grains, up to 100 μm in diameter, also occur rarely in each sample examined. These larger grains are more euhedral and contain regularly oriented planar lamellae 1–5 μm wide of a highly reflective, optically anisotropic oxide hosted by a less reflective matrix. The matrix of these grains appears to consist primarily of magnetite (or titanomagnetite), although in many cases, sphene and/or quartz are visible. The lamellae are of hematite and appear to be oriented along {111} planes in the magnetite hosts.

Electron microprobe analysis was undertaken to confirm the microscopic observations mentioned above. Because the electron microprobe cannot discriminate between ferric and ferrous iron, magnetite must be distinguished from hematite by the higher oxygen content of the latter (not analyzable with the microprobe), which results in proportionately lower total iron (reported as FeO). Thus the total oxide analysis of stoichiometric magnetite should sum to 93.1%, while that for hematite should sum to 89.8%. On this basis, the smaller disseminated oxide grains were determined to be nearly pure hematite, containing 1–4 wt % TiO_2, as shown in Table 2. Reliable analysis of either host or lamellae in the larger grains proved impossible due to unreasonably high SiO_2, Al_2O_3, CaO, and/or Na_2O values in the analyses, presumably from minute silicate grains present as inclusions or replacement products.

The presence of magnetite in a sample from site 1 was confirmed by X ray diffraction (XRD) analysis. Lattice plane spacings calculated from the XRD data indicate a nearly pure magnetite having little ulvospinel component. Hematite was also detected in this sample by XRD.

Analytical Procedures

Core samples from 2 to 8 cm long and 2.5 cm. diameter were obtained with a diamond-tipped stainless steel tube powered by a portable gasoline engine. Cores were oriented with a magnetic compass fitted to an inclinometer. The cores were cut into specimens 2.2 cm in length and rinsed for 30 s in 10 M HCl, whereupon they were stored in a magnetically shielded room having an ambient field less than 200 nT for the remainder of the study.

After measuring the natural remanent magnetization (NRM) of each sample, pilot samples from each site were stepwise progressively demagnetized by both thermal and alternating field techniques to determine which method was most appropriate for a given site. In all cases, thermal demagnetization proved more effective in unblocking a soft Quaternary field component, and the remaining samples were demagnetized thermally. Between 15 and 19 demagnetization steps were made between 110° and 660°C for each specimen.

Pilot alternating field demagnetization was done using a tumbling demagnetizer at peak fields between 1.5 and 80 mT. Thermal demagnetization was performed in a noninductively wound electrical resistance furnace equipped with a feedback circuit and thermocouple temperature measurement accurate to ± 5°C. Residual fields in the heating and cooling chambers were less than 2 nT. Sample magnetizations were measured using a three-axis cryogenic magnetometer with on-line data acquisition.

TABLE 2. Representative Microprobe Analyses

Weight Percent Oxide	Site	11A Hematite	11B Hematite	1 Hematite	11A Sphene
MgO		00.02	00.09	00.00	00.00
V_2O_3		00.19	00.24	00.03	00.38
TiO_2		03.15	02.81	02.74	37.01
MnO		00.15	00.11	00.17	00.02
SiO_2		00.46	00.21	00.34	31.01
FeO*		82.58	83.37	83.16	01.02
CaO		00.00	00.05	00.05	27.72
Al_2O_3		00.62	00.84	00.16	02.09
Total		87.17	87.71	86.62	99.25

FeO* is total Fe as FeO.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

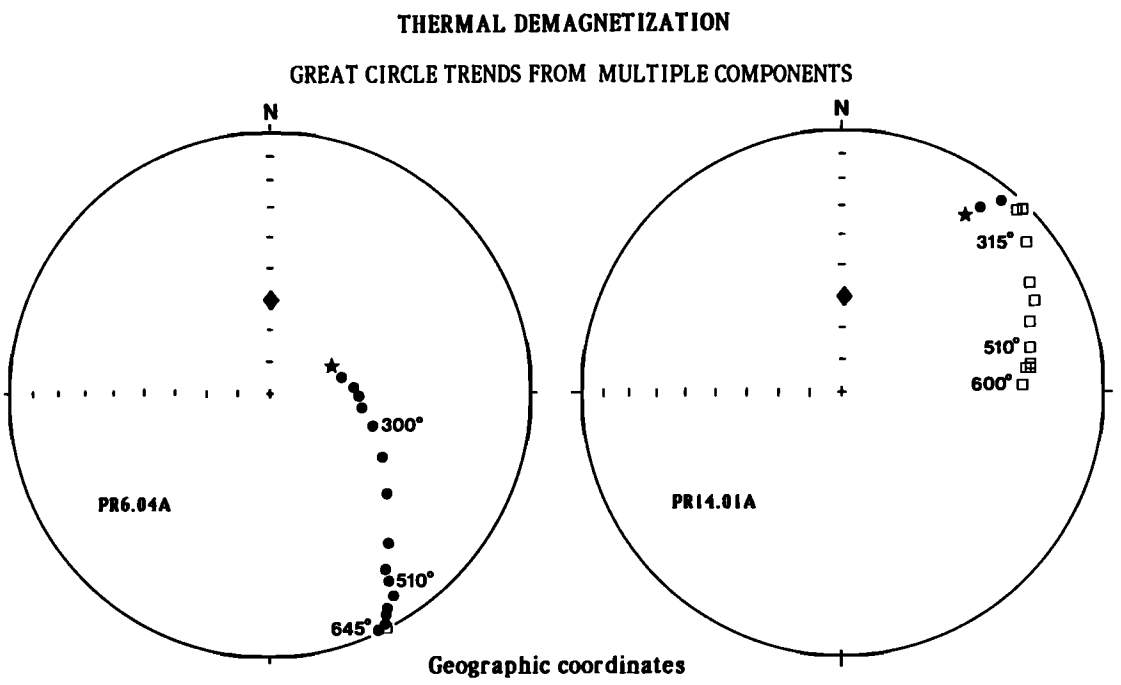


Fig. 3. Equal-area projection of progressively demagnetized remanence vectors for representative specimens (indicated by specimen number). Solid circles are directions in the lower hemisphere; open squares are in the upper hemisphere. Solid diamonds are the present axial dipole field direction. Stars are NRM directions.

PALEOMAGNETIC ANALYSIS

Demagnetization Behavior

NRM intensities of the tuffs range from 10^{-1} to 10^{-2} A/m. NRM directions of these samples are streaked between a steep northerly orientation and a more shallow, easterly to southeasterly direction. The mudstones (sites 12 and 13) are more weakly magnetized (NRM = 10^{-3} to 10^{-4} A/m) and move only 10°–30° from the Quaternary field direction during demagnetization before becoming unstable. Because these latter two sites are so heavily overprinted that pre-Quaternary directions could not be isolated, they were excluded from further consideration. Low-temperature demagnetization steps (up to 210°–260°C) on specimens from the remaining sites removes a component that (in geographic coordinates) is indistinguishable from the mean Quaternary normal direction. The direction and unblocking temperature spectrum of this component suggest growth of Fe-hydroxides in the weathering environment.

From 250° to 660°C, 15–17 demagnetization steps were made on every specimen. Viewed in equal-area projection, this detailed demagnetization record consistently revealed a great circle distribution (remagnetization circle) of directions (see Figures 3 and 4) for every specimen, indicating two components of magnetization. Most sites (except 14) had a pronounced drop in intensity over a narrow temperature range around 575°C (see Figure 5), apparently due to the unblocking of Ti-poor magnetite.

Above 590°C (hematite only), the directional behavior became variable from sample to sample, with two general types of movement: (1) some samples show no directional change and (2) most had a directional change back into the great circle trend already traversed. Some samples displaying this latter behavior had directions which moved farther along the great circle trend than had occurred at lower temperatures. The demagnetization behavior of these samples indicates the existence of magnetite in five of the six sites under consideration. In addition, two generations of hematite are present in all sites: one carries the same direction as does magnetite; the other is distinct and northward.

Magnetite Component

The direction carried by magnetite was isolated by least squares analysis [*Kirschvink*, 1980] of the linear trends of the demagnetized vectors' endpoints through the range 540°–600°C. Within this range, magnetization vectors decayed nearly linearly in orthogonal diagrams (see Figures 6*a* and 6*b*), indicating the removal of a single component of magnetization. The directions calculated for each specimen from a given site were pooled to calculate site mean directions, shown in Table 1.

A fold test, depicted in Figure 7, is obtained by comparison of the clustering of site mean directions before and after application of the structural correction, i.e., by examining the data in geographic and stratigraphic coordinates, respectively. The concentration parameter k is increased 4.5-fold when the samples are restored to paleohorizontal, indicating a prefolding magnetization. In stratigraphic coordinates the magnetite site means (Table 1) are well clustered about the magnetite formation mean (vector mean of all site means) direction at $D = 156.4°$, $I = -29.7°$, having $k = 101.4$ and $\alpha_{95} = 7.6°$. Statistical parameters are those of *Fisher* [1953].

The structual correction applied to these data involves rotation of bedding to paleohorizontal about a horizontal axis (the strike line). Such practice is strictly valid only if the fold(s) from which samples are collected are cylindrical and have horizontal axes; otherwise, errors manifested as spurious declinations may be introduced especially for plunges greater than

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

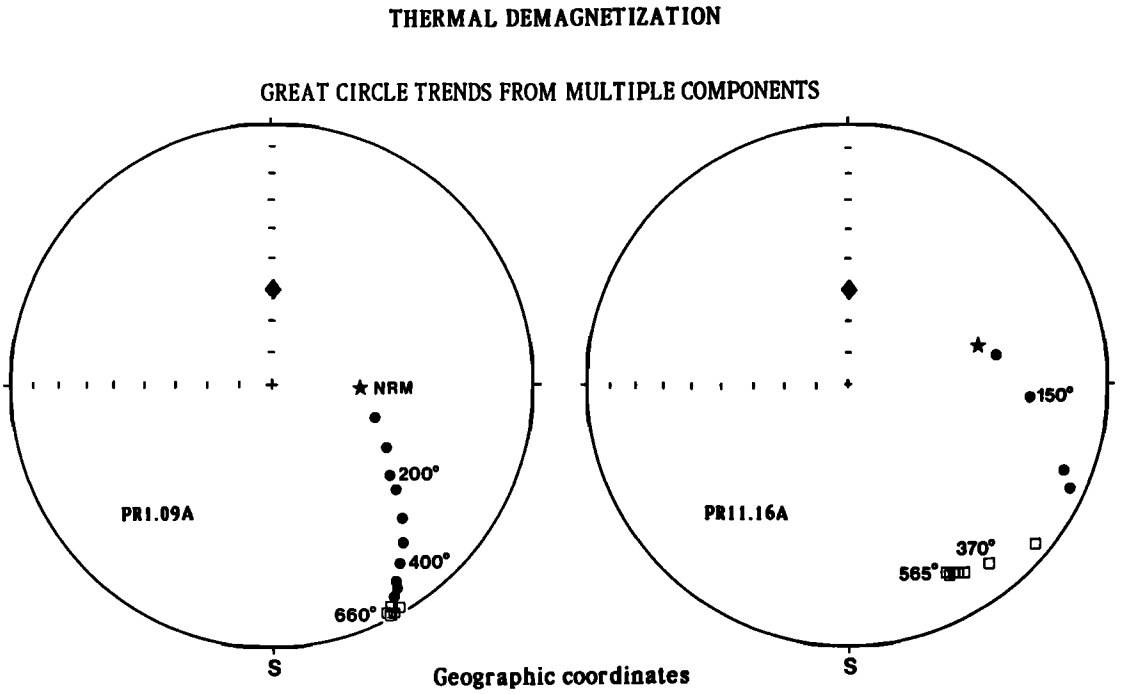


Fig. 4. Same conventions as Figure 3.

10° [*MacDonald*, 1980; *Scott*, 1984]. Folds plunging more than 15° have not been observed in the vicinity of the sample sites, and no evidence for systematic plunge exists in the region. In this case, the use of strike-line unfolding for dips less than 60° could introduce only minor error. Another potential problem is that dispersion resulting from uncertainty in the structural correction is magnified for large dips [*Renne et al.*, 1987, also manuscript in preparation, 1988]. Because site 7, with a dip of 82°, could introduce large errors, a formation mean direction omitting this site was also calculated. Excluding site 7 from calculation of the magnetite direction results in a formation mean having $D = 155.9°$, $I = -28.7°$, $k = 103.4$, indistinguishable within confidence limits from the direction computed from all five magnetite-bearing sites.

Hematite Components

As discussed above, one of the two directions carried by hematite is identical to the magnetite direction for each site and was therefore acquired before folding. The second hematite direction lies somewhere along the great circle trend discussed previously. Since the orientation of this remagnetization circle differs for each site (because each site has a different structural attitude), a postfolding origin is indicated for the second hematite direction. Indeed, the remagnetization planes viewed in geographic coordinates show a well-defined intersection (Figure 8) at a northerly downward direction, indicating that this direction is common to all sites and was acquired after folding. Similarly, the remagnetization planes intersect in stratigraphic coordinates (Figure 9) at a southesterly upward direction, coincident with the prefolding magnetite direction.

Analysis of Remagnetization Planes

Analysis of remagnetization circles has been dicussed by *Halls* [1976], *Hoffman and Day* [1978], *Onstott* [1980], and *Kirschvink* [1980]. This study used computer programs based on the methods of *Kirschvink* [1980] and *Onstott* [1980] to calculate the intersection directions and attendant statistics. A brief discussion of the algorithm follows.

The basic approach is to determine the direction perpendicular to each site's remagnetization plane, which is simply the pole to a great circle. Herein we use the term site mean "axle" for this pole to avoid confusion with paleomagnetic

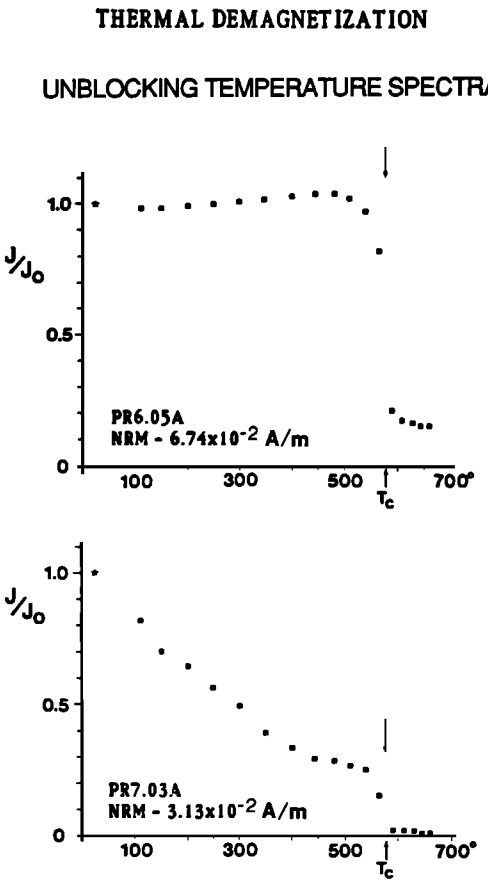


Fig. 5. Normalized intensity versus demagnetization temperature (in degrees Celsius) for two magnetite-bearing specimens. Tc is the Curie temperature of pure magnetite, shown for reference only.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

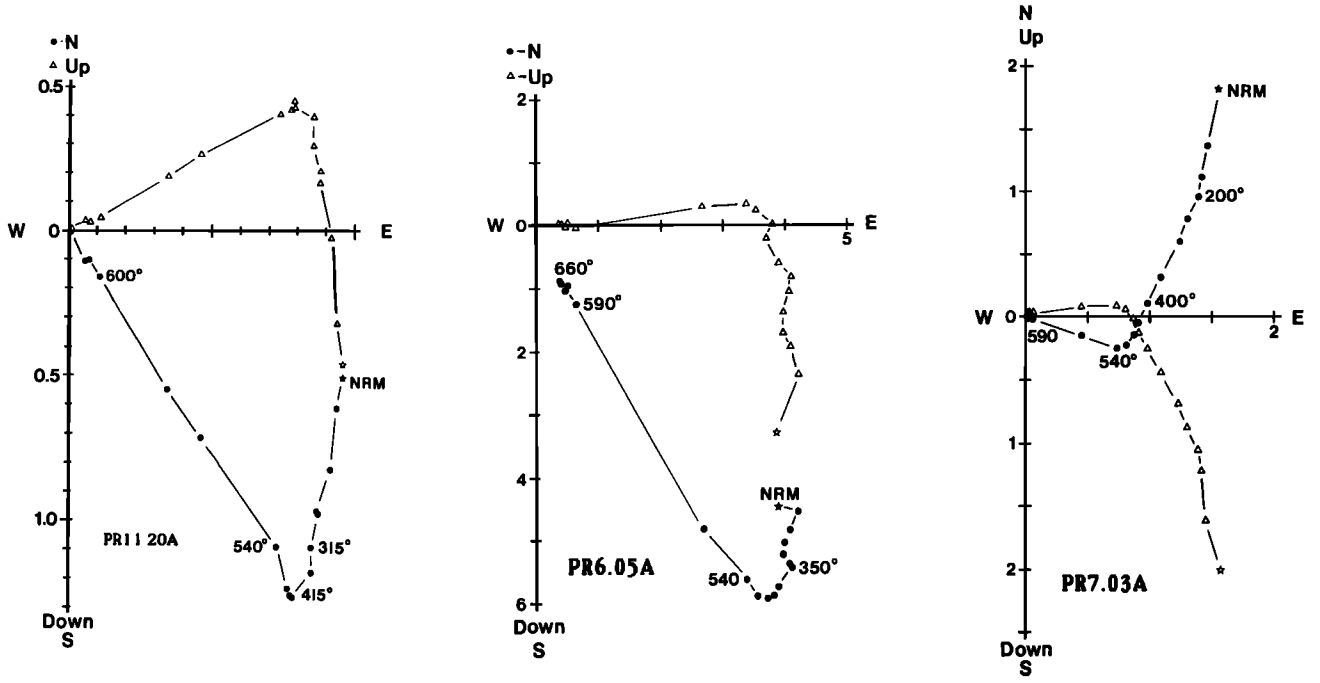


Fig. 6. Orthogonal demagnetization diagrams [*As and Zijderfeld*, 1958] for the specimens shown in Figure 5. Units are 0.01 A/m for PR6.05A and PR7.03A; 0.1 A/m for PR11.20A.

poles and to facilitate discussion. Site mean axles for two sites are shown in Figures 10 and 11. The axle for a given site is the direction orthogonal to both magnetization directions for that site. A direction which is perpendicular to each site's axle is the "common axle" and is calculated as the pole to a great circle fit to the individual site mean axles using Bingham parameters [*Onstott*, 1980]. The common axle is thus orthogonal to all the directions that are orthogonal to both magnetization components; hence it is the intersection of all sites' remagnetization planes. The common axle determined from data in geographic coordinates (Figure 12) is the postfolding direction and has $D = 352.3°$, $I = 35.0°$. The 95% confidence region obtained using the analysis of *Onstott* [1980] is an ellipse, reflecting asymmetric distribution of axles along a great circle, whose major axes $\alpha_{95}(1)$ and $\alpha_{95}(2)$ are 6.3° and 2.9°, respectively. The common axle in stratigraphic coordinates (Figure 13) is the prefolding direction and has $D = 160.3°$, $I = -31.5°$, $\alpha_{95}(1) = 7.0°$, $\alpha_{95}(2) = 3.6°$. Excluding site 7 (as discussed

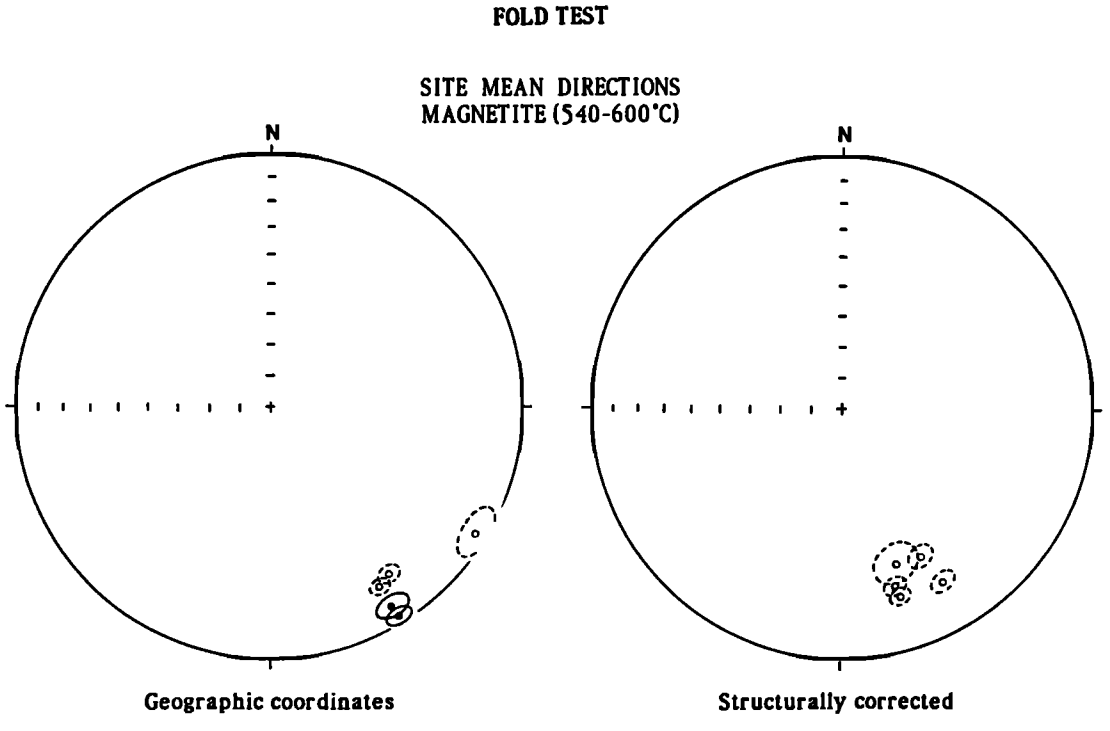


Fig. 7. Equal-area projections of magnetite directions (shown with α_{95} confidence cones) before and after structural correction. Solid boundaries on confidence cones denote lower hemisphere, dashed boundaries indicate upper hemisphere.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

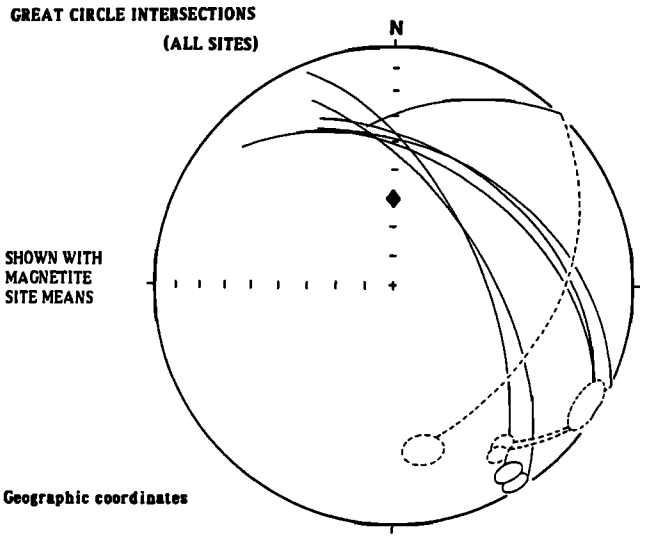


Fig. 8. Intersection of two-component remagnetization circles in geographic coordinates. Conventions as in Figure 7. Ovals are magnetite site mean α_{95}, shown for reference. Site 14, lacking magnetite, is shown with the formation magnetite mean restored to site 14 geographic coordinates.

above) yields a mean having $D = 154.6°$, $I = -29.4°$, $\alpha_{95}(1) = 8.9°$, $\alpha_{95}(2) = 3.0°$

The first step in this analysis was to calculate the mean of the sample axles for each site. A noteworthy feature of these axle distributions is the tendency for some specimen axles to be displaced toward the Quaternary directed axle, which is defined as the perpendicular to the great circle connecting the Quaternary direction with the magnetite mean for that site. About 25% of the specimen axles for each site were at or near the site Quaternary axle and were rejected from further calculations. This is apparently caused by a high blocking temperature viscous remanence and/or hematite growth in the weathering environment, resulting in a Quaternary magnetization that completely overprints the earlier postfolding magnetization.

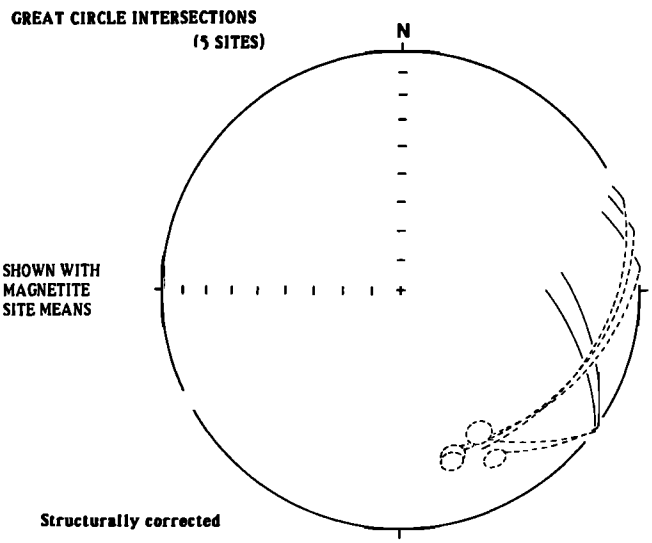


Fig. 9. Intersection of remagnetization circles in stratigraphic coordinates. Same conventions as Figure 8. Great circle without a magnetite mean is from site 14. Site 7 is omitted (see text).

Analysis of the demagnetization trend was done in two different ways (as discussed by *Kirschvink* [1980]) that were found to be generally comparable. For each sample a great circle was calculated over the temperature range of maximum directional change (250°–510°C). A plane (defined by the distribution about a center of mass of vector endpoints weighted by intensity) was also calculated over the temperature range of maximum vector displacement (250°–660°C). Both techniques had high precision and yielded means within 2° of each other, indicating that the data used in the calculations were effectively cleaned of all but two magnetization directions. As no justifiable preference exists for either of these techniques, their results were averaged (Table 3). An exception was the site without a magnetite component (site 14), where the planes technique had a much higher error (although a mean within 3° of the great circle result). Only the great circle results were used for site 14.

Oxide Paragenesis and Timing of Remanence Acquisition

In order to draw tectonic conclusions from paleomagnetic data, the various components of remanent magnetization must be dated (or at least constrained in age) and, ideally, correlated with independently recognized geologic events. In the present case, three components (neglecting the Quaternary weathering overprint) must be considered: (1) a prefolding component identified with hematite, (2) a prefolding component due to Ti-poor magnetite, and (3) a postfolding component associated with hematite.

Magnetite

The existence of magnetite in samples of this study has been discussed above and is consistent with the pronounced intensity drop during thermal demagnetization of most specimens over a very discrete temperature range centered at about 570°C. Thus the high-intensity RM component has a laboratory unblocking temperature $T_{ub} \approx 570°C$. Such a high T_{ub} implies a blocking temperature $T_b \approx T_{ub}$ [*Pullaiah et al.*, 1975]. This blocking temperature is very near the Curie temperature (T_c) of pure magnetite, and the magnetite responsible for this component therefore contains less than 5 mol % ulvospinel (Usp) [*O'Reilly*, 1984]. This is somewhat more magnetite-rich than microphenophenocrysts of titanomagnetite that are large enough ($> \sim 5$ μm) to be analyzed by electon microprobe in typical felsic ignimbrites; these average about 10–60 mol % Usp, depending on equilibration temperature and oxygen fugacity [*Buddington and Lindsley*, 1964; *Reynolds*, 1977]. Nonetheless, titanomagnetite laboratory unblocking temperatures of 560°–580°C determined for Nosoni tuffs are consistent with results from many samples of Cenozoic felsic ash flow tuffs [e.g., *Gose*, 1971; *Reynolds*, 1977; *Geissman et al.*, 1982; *Rosenbaum*, 1986].

Exsolution of the cubic solid solution into magnetite-rich and ulvospinel-rich phases can occur if the posteruptive cooling rate is slow enough to permit unmixing. Protracted cooling of the Skookum Tuff (sites 1, 6, and 7) is evidenced by extensive subsolidus high-temperature devitrification and welding, and the latter feature is well developed in the tuff of site 11A: Thus favorable conditions for exsolution probably occurred. Unmixing will be most complete in smaller grains, whose diffusive volumes are small. This may explain why submicrometer-sized single-domain or pseudo-single-domain titanomagnetite grains, the main contributor to thermorema-

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

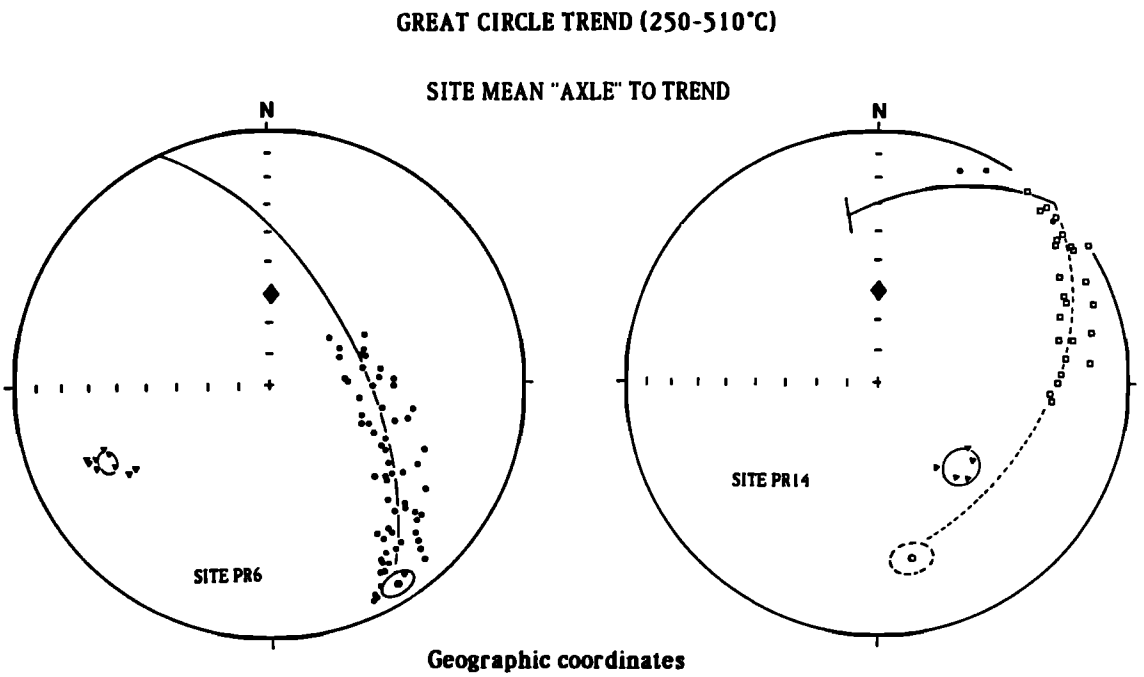


Fig. 10. Directions in geographic coordinates between 250° and 510°C for specimens from sites 6 and 14. Symbols and projection as in Figure 3. Great circles are least squares best fits to data. Triangles are specimen axles, shown with Fisher α_{95} confidence cones. Site mean magnetite directions shown for reference.

nent magnetization (TRM) [*Dunlop*, 1973], almost invariably have laboratory unblocking temperatures corresponding to nearly pure magnetite in ash flow tuffs, while larger grains often contain appreciable Usp.

The prefolding magnetite direction is interpreted to be a product of thermoremanence acquired during initial postemplacement cooling of the pyroclastic rocks during the Guadelupian Stage of the Permian Period. This prefolding direction is always reversed, consistent with acquisition during the Kiaman [*Irving and Parry*, 1963; *McMahon and Strangway*, 1968] long reversed polarity superchron of the geomagnetic field.

The survivability of magnetite-carried magnetization through burial metamorphism must be considered in addressing the acquisition age of this component. It is well established that the unblocking of remanent magnetization is time and temperature dependent; according to the analysis given by

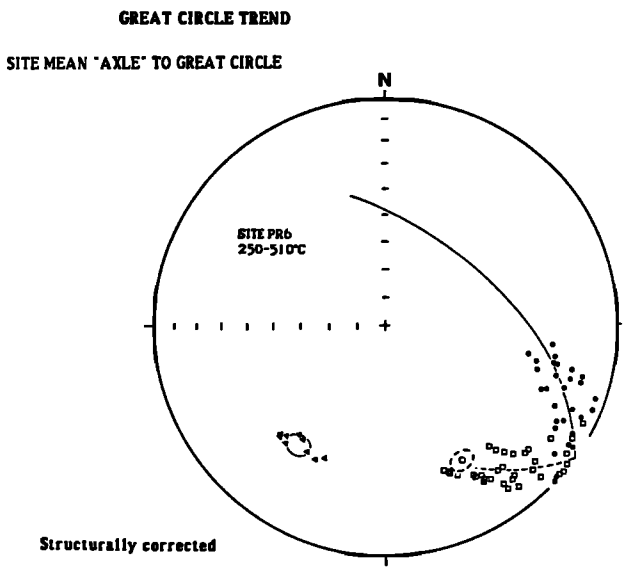


Fig. 11. Same explanation as Figure 10 except shown in stratigraphic coordinates.

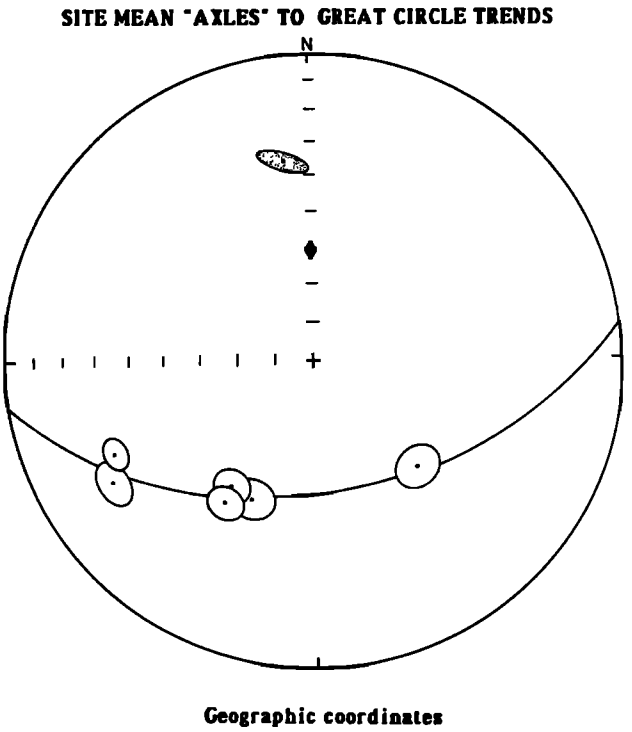


Fig. 12. Site mean axles with least least squares best fit great circle. Common axle is defined as the pole to this great circle, shown with confidence ellipse (shaded). Geographic coordinates.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

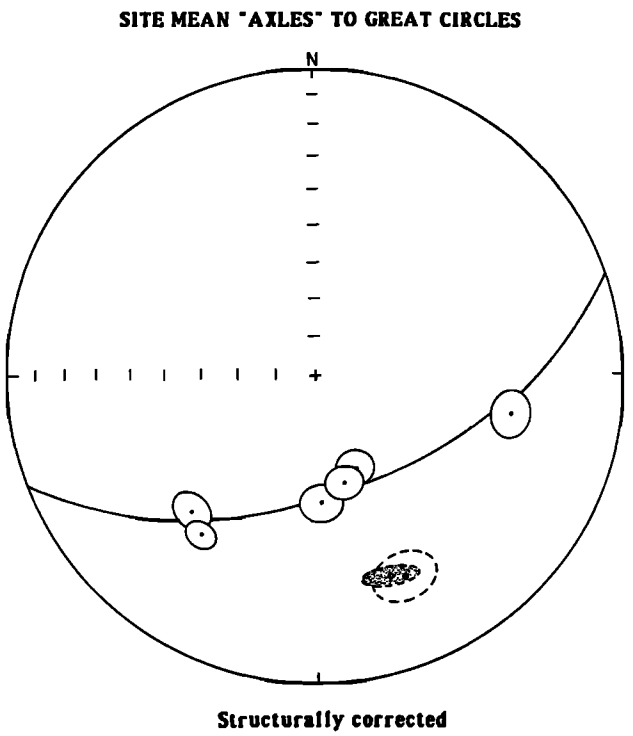


Fig. 13. Same explanation as Figure 12 but in stratigraphic coordinates. Formation magnetite mean with its α_{95} ellipse shown for comparison with common axle.

Pullaiah et al. [1975], magnetite with a blocking temperature higher than 550°C can be unblocked only at temperatures above 500°C, even if this temperature is maintained for 4.5 b.y. As discussed above, the magnetite components isolated in this study have blocking temperatures above 550°C and are thus expected to survive heating to 325°C (maximum estimated peak metamorphic temperature) even if this temperature were maintained for 260 m.y., the approximate stratigraphic age of the Nosoni Formation.

Hematite

Hematite, ubiquitous in samples from all sites, is likely to have formed under two sets of circumstances. The depositional environments of all sampled lithologies are shallow water to subaerial, hence oxidizing by nature. Hematite is ubiquitous in many ash flow tuffs and is an important contributor to their paleomagnetic remanence [*Gose*, 1971; *Nairn et al.*, 1975; *Reynolds*, 1977; *Geissman et al.*, 1982]. Hematite is produced both during high-temperature devitrification and subsequent weathering [*Nairn et al.*, 1975; *Reynolds*, 1977] of tuffs, usually soon after eruption, and the directional component carried by this phase tends to average out secular variation of the geomagnetic field. These are probable parageneses for some of the hematite present in the Nosoni tuffs and would result in a hematite-borne paleomagnetic direction essentially indistinguishable from the magnetite TRM direction. Additional growth of hematite during metamorphism at a significantly later time, accompanying the crystallization of sphene and albite, is discussed below.

Sphene is notably associated with hematite within the larger opaque grains in all the polished sections examined and is probably an alteration product of the exsolved ulvospinel-rich phase, accomodating Ti and minor Fe (see microprobe analysis Table 2). The albitization of originally more calcic plagioclase, which probably occurred during peak metamorphism in the Early to Middle Jurassic [*Renne and Scott*, 1988], provides a source of Ca for the crystallization of sphene. A by-product of this reaction under sufficiently high oxygen fugacity is hematite:

$$\underset{\text{(from plag)}}{Ca^{2+}} + \underset{\text{(ulvospinel)}}{Fe_2TiO_4} + \underset{\text{(quartz)}}{SiO_2} + O_2 = \underset{\text{(sphene)}}{CaTiSiO_5} + \underset{\text{(hematite)}}{Fe_2O_3}$$

Thus two stages of hematite growth, one essentially primary (Permian) and the other during metamorphism (Early to Middle Jurassic), are consistent with the prefolding and post-

TABLE 3. Axle Data

Site	Temperature Range, °C		Geographic Coordinates		Axle Statistics			Stratigraphic Coordinates	
	Great Circles	Planes	Declination	Inclination	N	α_{95}	k	Declination	Inclination
1	250–510	250–660	239.2	24.6	9	5.3	94	222.4	39.9
6	250–510	250–660	245.3	29.8	9	3.9	177	216.1	36.0
7	250–510	250–660	204.7	48.9	7	5.9	104	101.5	35.4
11A	260–510	260–660	212.7	44.6	6	4.7	201	165.3	60.1
11B	260–510	260–660	213.7	49.2	7	4.8	160	157.4	63.0
14	260–510		136.2	50.1	6	5.7	139	176.9	55.5
	common axle		352.3	35.0				160.3	−31.5

	Common Axle Bingham Statistics	
	Geographic	Stratigraphic
$k1$	−243.0	−235.2
$k2$	−3.5	−5.3
α_{95} (1)	6.3	8.9
α_{95} (2)	2.9	3.0
Oval azimuth	59.5	68.6

Site 14 has dip = 30°, dip direction = 256°. $k1$, $k2$, α_{95} (1), and α_{95} (2) are Bingham statistical parameters described by *Onstott* [1980]. Oval azimuth is the angle between azimuth of mean direction and major axis of error ellipse.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

folding directions of magnetic remanence carried by this phase.

The consistency of paleomagnetic directions in ash flow tuffs have been shown to be somewhat variable, depending to some extent on vertical position. Differential compaction, variable degrees of welding and devitrification, and zonally variable oxidation probably all contribute to variation of rock magnetic properties and directional consistency of ash flow tuffs [*Nairn et al.*, 1975; *Reynolds*, 1977; *Geissman*, 1980; *Geissman et al.*, 1982; *Rosenbaum*, 1986]. *Reynolds* [1977] and *Geissman* [1980] found that directional consistency was highest in basal vitrophyres, while *Nairn et al.* [1975] found greater clustering in the interior portions of ash flow tuffs. *Rosenbaum* [1986] found that either basal or interior portions of rhyolitic ash flow tuffs may be subject to dispersion of directions due to welding at temperatures below the blocking temperature of magnetic minerals. No systematic stratigraphic variations in magnetic properties or orientation of paleomagnetic directions in Nosoni tuffs were observed in this study although the exposed stratigraphic extent of each outcrop was sampled. Lack of dispersion of magnetite-carried directions in the welded Nosoni tuffs indicates that compaction during welding was complete at temperatures above 570°C. The relatively high eruption temperature and low viscosity implied for silica-poor ignimbrites (i.e., lacking quartz phenocrysts, as those of this study) may promote rapid posteruptive compaction that is completed at temperatures above 570°C.

Secular Variation

Quickly cooled volcanic rocks (flows and tuffs) yield paleomagnetic directions whose corresponding poles may not coincide with the earth's rotation axis because of secular variation. Directions carried by magnetite are the most likely to show variation because of the geologically rapid time interval of remanence acquisition by this phase upon initial posteruptive cooling. Little or no secular variation is evident in magnetite directions from these Nosoni sites, as directions from sites spanning at least 200 m of section are well clustered within individual sites (k = 79–274) and between sites (k = 101). This is consistent with the conclusion of *Irving and Pullaiah* [1976] that the magnitude of secular variation is at a minimum during long, stable polarity chrons such as the Kiaman, during which the Nosoni Formation was deposited. The postulated origins of hematite both yield chemical remanent magnetization (CRM), whose acquisition probably occurred over a sufficiently long time interval that secular variation is averaged out.

COMPARISON WITH REFERENCE DATA

In the foregoing, a primary (Permian) and a secondary (Jurassic) direction of magnetization have been identified in samples from sites in the Nosoni Formation. The pole corresponding to the Permian direction (magnetite formation mean) is located at 57.6°N latitude, 103.0°E longitude. This pole falls on the North American apparent polar wander (APW) path given by *Irving and Irving* [1982] (Figure 14), and coincides closely to Upper Permian cratonic poles. As the 95% confidence intervals overlap, no statistically significant difference can be inferred between the mean Permian pole for the sample sites and that for the craton. Comparison with the Guadelupian reference pole (calculated using a 30-m.y. window) given by *Irving and Irving* [1982] suggests 7° of apparent clockwise rotation of the sites with respect to the craton, which is not statistically significant.

The paleomagnetic pole calculated from the postfolding overprint direction, which is interpreted to have been acquired in the Early to Middle Jurassic, is located at 67.0°N, 104.1°E. This pole intercepts the APW path for the North American craton, coinciding with middle Early Jurassic poles given by *May and Butler* [1986] (Figure 15). A host of well-dated reference poles are available for this time interval (in contrast to the Late Permian), and because the North American craton was moving relatively fast in the Early to Middle Jurassic, the APW path given by *May and Butler* [1986] permits high temporal resolution of poles. The closest correspondence is with a pole determined by *Smith and Noltimier* [1979] from 179 Ma [*Sutter and Smith*, 1979] Newark Trend Group 2 igneous rocks in New England. Substantial overlap of 95% confidence intervals also occurs with a pole determined by *May et al.* [1986] from 171 Ma ash flow tuffs in southeastern Arizona. Correction of the Nosoni postfolding pole for 7° clockwise rotation suggested by slight discordance of the Permian pole brings the former into excellent coincidence with the *May and Butler* [1986] 171 Ma pole and is consistent with the isotopic (pre-168 Ma) and biostratigraphic (post-Bajocian) constraints on the age of folding in the RS.

DISCUSSION AND CONCLUSIONS

The two directions of magnetization delineated in this study have paleomagnetic poles whose positions correspond within error to the positions of appropriately aged reference poles for the North American craton. These data indicate that neither significant latitudinal transport nor rotation of the sampling region has occurred with respect to the craton since the Permian. This conclusion bears some discussion, because other investigators have reported clockwise rotations in the RS of 60°–116° based on paleomagnetic data. No statistically significant latitudinal discordance (anomalous inclination) has been reported.

The interpretation of tectonic rotations in the Klamath Mountains inferred from paleomagnetic data must consider the possibility of oroclinal bending [*Carey*, 1955]. Differential rotations between different structural domains in the RS are expected in view of the 150° variation in the trends of F_1 structures [*Renne and Scott*, 1986, 1988; *Renne et al.*, 1986*a*, *b*]. Some of the paleomagnetically inferred rotations reported from the RS [*Fagin and Gose*, 1983; *Achache et al.*, 1982; *Mankinen et al.*, 1984] may be easily explained in this context, as discussed at length by *Renne and Scott* [1986, 1988] wherein the different structural domains of the RS are outlined. Some paleomagnetic data (from sites shown in Figure 2) purported to display rotation, however, have been reported from the same structural domain (herein termed the Nosoni Creek domain) occupied by the sites of this study. In order to resolve or at least clarify the apparently conflicting results, a brief review of previous work is desireable.

Previous Results From the Nosoni Creek Structural Domain

Fagin and Gose [1983] report data from two sites within the Nosoni Creek domain which they interpret to show large rotations: one site is in the McCloud Limestone; the other is in the Nosoni Formation. Very little information bearing on the accuracy of their data is presented by *Fagin and Gose* [1983], though slightly more detailed discussion is given by *Fagin*

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

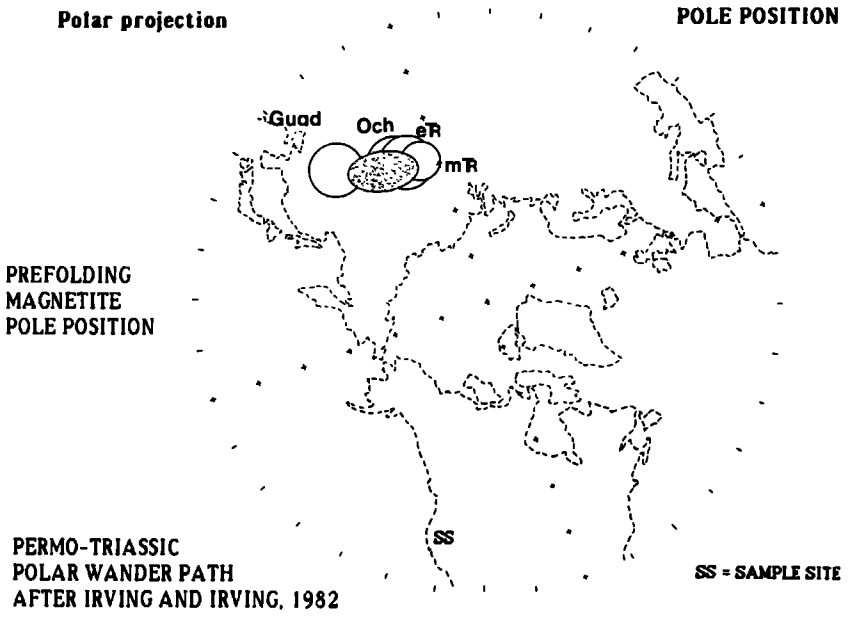


Fig. 14. Comparison of formation mean magnetite pole with Middle Permian through Middle Triassic APW path for North American craton [after *Irving and Irving*, 1982]. Shaded ellipse is 95% confidence region.

[1983]. No fold test is applicable to these sites; the timing of magnetization is therefore conjectural. Their specimens were demagnetized in six steps or fewer, which (based upon this study) is insufficient to allow the recognition or isolation of all the components of magnetization. The Nosoni Formation site of *Fagin and Gose* [1983] is reported by *Fagin* [1983] as occurring in a basalt at a location where no lava flows exist in the Nosoni; the only rocks resembling basalts here are mafic dikes whose ages are unknown. The directions reported for this site are poorly clustered ($k = 7$), and the site mean (in geographic coordinates) cannot be statistically distinguished from the present field.

Sites sampled by *Mankinen et al.* [1984] from a "Nosoni Creek transect" also occupy this same structural domain, as shown in Figure 2. Paleomagnetic directions from two of these sites are depicted by *Mankinen et al.* [1984] in their Figure 2 and are interpreted by Mankinen et al. as primary Permian directions that have been tectonically rotated approximately 97° clockwise. Few data are presented by *Mankinen et al.* [1984] that permit evaluation of this conclusion. Their two sites from the Nosoni Creek domain have directions that are very close to the Quaternary normal field direction before structural correction, and no evidence (e.g., demagnetization data) is given to show that these directions are not overprinted by the Quaternary field. Furthermore, these two sites appear to fail a fold test.

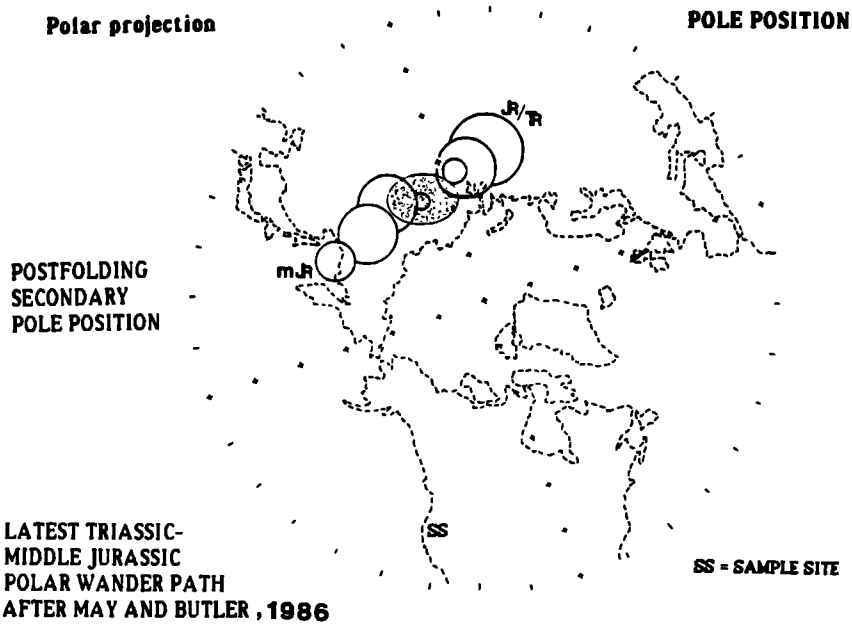


Fig. 15. Comparison of postfolding common axle pole with Jura-Triassic through Middle Jurassic APW path from *May and Butler* [1986]. Confidence region (shaded ellipse) corresponds to maximum dimension in error ellipse of paleomagnetic direction.

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The prefolding magnetization directions reported herein are likely to be primary because they pass a fold test (hence are demonstrably pre-Middle Jurassic), are carried in most sites by magnetite, are reversed as expected, and are distinguishable from the Quaternary field in geographic coordinates. On such criteria, these data must be considered the most reliably primary directions yet reported from the Nosoni Creek domain.

Tectonic Rotation of Other Domains

Paleomagnetic data showing large rotations from elsewhere in the RS appear more credible. Detailed discussion of the significance of these rotations and their structural context is presented by *Renne and Scott* [1988]. Here we intend simply to observe that although the Nosoni Creek domain appears unrotated, paleomagnetic evidence does exist for tectonic rotations elsewhere in the RS. Directions from the Hawkins Creek domain ("Hawkins Creek Transect") reported from Permian and Triassic units (including the Nosoni Formation) by *Mankinen et al.* [1984] are reversed and pass a fold test, permissible evidence of a primary origin. Approximately 100° of post-Permo Triassic clockwise rotation with respect to the craton is indicated by these directions. *Fagin and Gose* [1983] report data from three sites in the Hawkins Creek domain which are consistent with results of *Mankinen et al.* [1984]. Unfortunately, no evidence can be garnered from *Fagin and Gose* [1983] or *Fagin* [1983] that their reported directions are primary.

Achache et al. [1982] reported demonstrably postfolding (hence presumably post-early Middle Jurassic) reversed polarity remagnetizations from the Devonian Kennett Limestone in the Backbone domain. *Achache et al.* [1982] propose that remagnetization accompanied the intrusion of the nearby 140 Ma Shasta Bally batholith and conclude that approximately 60° of clockwise rotation has occurred since this time. Although their postulated timing of remagnetization is open to question, substantial clockwise rotation of their sample sites is indicated by any post-Middle Jurassic age of remagnetization.

Paleomagnetic data from the RS which meet reliability criteria outlined above clearly show differential clockwise rotations of up to approximately 100°. Studies from three separate structural domains show systematic rotations whose magnitudes are clearly distinct within the errors cited by the various authors. The RS therefore cannot be considered a uniformly rotated entity, as has been suggested by *Fagin and Gose* [1983], *Mankinen et al.* [1984], and *Bogen* [1986]. Oroclinal bending appears to be the most reasonable explanation for differential rotations in the RS and in the Klamath Mountains as a whole [*Renne and Scott*, 1986, 1988]. Further paleomagnetic research (in progress) in the RS will help clarify this issue. The absence of anomalous inclinations in any paleomagnetic data from the RS, including those of this study, indicates that the RS has shared a common latitudinal transport history with cratonic North America since at least Permian time.

Acknowledgments. Field expenses were partly defrayed by two Geological Society of America Penrose Grants to P.R.R.; all other expenses were borne by the authors. Z-Axis Exploration, Inc. of Pleasant Hill, California, generously donated the use of their superb paleomagnetism laboratory and sampling equipment. J. M. Johnston of Z-Axis helped collect samples, made many of the measurements reported herein, and helped interpret the data in early stages of the study. Field assistance by J. M. Johnston, Q. A. Becker, J. J. Irwin, T. Kharadyer, M. M. Fulford, and L. Pelloran is appreciated. E. A. Mankinen and W. P. Irwin have been generous with their time and in sharing unpublished data; discussions with them have been invaluable to us. S. L. Gillett, R. B. Hargraves, and T. C. Onstott provided computer programs (and instruction in the use thereof) used for data analysis. Reviews of the manuscript by T. C. Onstott and JGR reviewers E. A. Mankinen and J. W. Geissman are appreciated. We thank J. E. Wright for our geological introduction to the Klamath Mountains. P.R.R. gratefully acknowledges the encouragement of the late John P. Albers.

REFERENCES

Achache, J., A. Cox, and S. O'Hare, Paleomagnetism of the Devonian Kennett Limestone and the rotation of the eastern Klamath Mountains, California, *Earth Planet. Sci. Lett.*, *61*, 365–380, 1982.

Albers, J. P., and J. F. Robertson, Geology and ore deposits of East Shasta copper-zinc district, Shasta County, California, *U.S. Geol. Surv. Prof. Pap.*, *388*, 1961.

As, J. A., and J. D. A. Zijderfeld, Magnetic cleaning of rocks in paleomagnetic research, *Geophys. J. R. Astron. Soc.*, *1*, 308–319, 1958.

Bogen, N. L., Paleomagnetism of the Upper Jurassic Galice Formation, southwestern Oregon: Evidence for differential rotation of the eastern and western Klamath Mountains, *Geology*, *14*, 335–338, 1986.

Buddington, A. F., and D. H. Lindsley, Iron-titanium oxide minerals and synthetic equivalents, *J. Petrol.*, *5*, 310–357, 1964.

Carey, S. W., The Orocline concept in geotectonics, 1, *Proc. R. Soc. Tasmania*, *89*, 255–288, 1955.

Coney, P. J., D. L. Jones, and J. W. H. Monger, Cordilleran suspect terranes, *Nature*, *288*, 329–333, 1980.

Coogan, A. H., Stratigraphy and paleontology of the Permian Nosoni and Dekkas formations (Bollibokka Group), *Univ. Calif. Berkeley Publ. Geol. Sci.*, *36*, 243–316, 1960.

Dunlop, D. J., Superparamagnetic and single-domain threshold sizes in magnetite, *J. Geophys. Res.*, *78*, 1780–1793, 1973.

Fagin, S. W., Paleogeography and tectonics of the Redding Section, eastern Klamath belt, northern California, Ph.D. dissertation, Univ. of Tex. at Austin, Austin, 1983.

Fagin, S. W., and W. A. Gose, Paleomagnetic data from the Redding Section of the eastern Klamath belt, northern California, *Geology*, *11*, 505–508, 1983.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295–305, 1953.

Geissman, J. W., Paleomagnetism of ash flow tuffs: Microanalytical recognition of TRM components, *J. Geophys. Res.*, *85*, 1487–1499, 1980.

Geissman, J. W., R. Van der Voo, and K. L. Howard, A paleomagnetic study of the structural deformation in the Yerington district, Nevada, *Am. J. Sci.*, *282*, 1042–1109, 1982.

Gose, W. A., Paleomagnetic studies of Miocene ignimbrites from Nevada, *Geophys. J. R. Astron. Soc.*, *20*, 241–252, 1971.

Halls, H. C., A least-squares method to find a remanence direction from converging remagnetization circles, *Geophys. J. R. Astron. Soc.*, *45*, 297–304, 1976.

Hoffman, K. A., and R. Day, Separation of multicomponent NRM: A general method, *Earth Planet. Sci. Lett.*, *40*, 433–438, 1978.

Irving, E., and G. A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv.*, *5*, 141–188, 1982.

Irving, E., and L. G. Parry, The magnetism of some Permian rocks from New South Wales, *Geophys. J. R. Astron. Soc.*, *7*, 395–411, 1963.

Irving, E., and G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, *Earth Sci. Rev.*, *12*, 35–64, 1976.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

LaPierre, H., F. Albarede, J. Albers, B. Cabanis, and C. Coulon, Early Devonian volcanism in the eastern Klamath Mountains, California: Evidence for an immature island arc, *Can. J. Earth Sci.*, *22*, 214–227, 1985.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, *J. Geophys. Res.*, *85*, 3659–3669, 1980.

Mankinen, E. A., W. P. Irwin, and C. S. Gromme, Implications of paleomagnetism for the tectonic history of the eastern Klamath and related terranes in California and Oregon, in *Geology of the Upper Cretaceous Hornbrook Formation, Oregon and California, Publ. 42*, edited by T. H. Nilson, pp. 221–229, Pacific Section, Society of

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Economic Paleontologists and Mineralogists, Bakersfield, Calif., 1984.

May, S. R., and R. F. Butler, North American Jurassic apparent polar wander: Implications for plate motion, paleogeography and Cordilleran tectonics, *J. Geophys. Res.*, *91*, 11,519–11,544, 1986.

May, S. R., R. F. Butler, M. Shafiqullah, and P. E. Damon, Paleomagnetism of Jurrassic volcanic rocks in the Patagonia Mountains, southeastern Arizona: Implications for the North American 170 Ma reference pole, *J. Geophys. Res.*, *91*, 11,545–11,555, 1986.

McMahon, B. E., and D. W. Strangway, Stratigraphic implications of paleomagnetic data from upper Paleozoic-lower Triassic redbeds of Colorado, *Geol. Soc. Am. Bull.*, *79*, 417–428, 1968.

Murray, M., and K. C. Condie, Post-Ordovician to early Mesozoic history of the eastern Klamath subprovince, northern California, *J. Sediment. Petrol.*, *43*, 505–515, 1973.

Nairn, A. E. M., P. D. Rowley, and J. J. Anderson, Paleomagnetism of selected Tertiary volcanic units, southwest Utah, *Spec. Pap. Geol. Soc. Am.*, *160*, 63–74, 1975.

Onstott, T. C., Application of the Bingham distribution function in paleomagnetic studies, *J. Geophys. Res.*, *85*, 1500–1510, 1980.

O'Reilly, W., *Rock and Mineral Magnetism*, 220 pp., Blackie, Glasgow, Scotland, 1984.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett.*, *28*, 133–143, 1975.

Renne, P. R., and G. R. Scott, Comment on "Paleomagnetism of the upper Jurassic Galice Formation: Evidence for differential rotation of the eastern and western Klamaths" by N. L. Bogen, *Geology*, *14*, 1048–1049, 1986.

Renne, P. R., and G. R. Scott, Constraints on the Permian through Jurassic tectonic evolution of the eastern Klamath Mountains (Redding Section), California, *Tectonics*, in press, 1988.

Renne, P. R., G. H. Curtis, and G. R. Scott, Nature and timing of deformation in the Redding Section, eastern Klamath Mountains, California, *Geol. Soc. Am. Abstr. Programs*, *18*(2), 175, 1986*a*.

Renne, P. R., G. R. Scott, and D. R. Bazard, Multicomponent paleomagnetic data from the Redding Section, eastern Klamath Mountains: Cratonic Permian directions with secondary middle Jurassic overprints, *Geol. Soc. Am. Abstr. Programs*, *18*(2), 175, 1986*b*.

Renne, P. R., G. R. Scott, and A. L. Deino, Structural correction and associated error in paleomagnetism (abstract), *Eos Trans. AGU*, *68*, 1251, 1987.

Reynolds, R. L., Paleomgnetism of welded tuffs of the Yellowstone Group, *J. Geophys. Res.*, *82*, 3677–3693, 1977.

Rosenbaum, J. G., Paleomagnetic directional dispersion produced by plastic deformation in a thick Miocene welded tuff, southern Nevada: Implications for welding temperatures, *J. Geophys. Res.*, *91*, 12,817–12,834, 1986.

Sanborn, A. F., Geology and paleontology of the southwest quarter of the Big Bend Quadrangle, Shasta County, California, *Spec. Rep. Calif. Div. Mines Geol.*, *63*, 26 pp., 1960.

Scott, G. R., Anomalous declinations from plunging structures (abstract), *Eos Trans. AGU*, *65*, 866, 1984.

Scott, G. R., P. R. Renne, D. R. Bazard, and J. M. Johnston, Paleomagnetism of a Permian Nosoni Fm. ignimbrite and implications to accretionery models of eastern Klamath Belt tectonics, *Geol. Soc. Am. Abstr. Programs*, *17*(6), 407 1985.

Skinner, J. W., and G. L. Wilde, Permian biostratigraphy and fusulinid faunas of the Shasta Lake area, northern California, *Harold N. Fisk Mem. Pap. Paleontol. Contrib. Protozoa 6*, 98 pp., Univ. of Kans., Lawrence, 1965.

Smith, T. E., and H. C. Noltimier, Paleomagnetism of the Newark trend igneous rocks of the north central Applachians and the opening of the central Atlantic Ocean, *Am. J. Sci.*, *279*, 778–807, 1979.

Speed, R. C., Collided Paleozoic microplate in the western United States, *J. Geol.*, *87*, 279–292, 1979.

Sutter, J. F., and T. E. Smith, $^{40}Ar^{39}/Ar$ ages of diabase intrusions from Newark trend basins in Connecticut and Maryland: Initiation of central Atlantic rifting, *Am. J. Sci.*, *279*, 801–831, 1979.

Turner, F. J., *Metamorphic Petrology*, 524 pp., McGraw-Hill, New York, 1981.

Wright, J. E., Permo-Triassic accretionery subduction complex, southwestern Klamath Mountains, northern California, *J. Geophys. Res.*, *87*, 3805–3818, 1982.

D. R. Bazard, Department of Geosciences, University of Arizona, Tucson, AZ 85721.

P. R. Renne, Department of Geological and Geophysical Sciences, Princeton University, Princeton, NJ 08544.

G. R. Scott, Lodestar Magnetics Inc., 1069 62nd St., Oakland, CA 94608.

(Received May 11, 1987;
revised November 25, 1987;
accepted December 18, 1987.)

21562202b, 1988, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB04p03387 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License